

Geometry: Unit 13, Lesson 10

1. Two line segments, $\overline{MP}$ and $\overline{CK}$, are located on the coordinate plane, such that $M(2,3)$, $P(6,1)$, $C(4,6)$, and $K(10,3)$. Why are $\overline{MP}$ and $\overline{CK}$ parallel segments?
2. Line segment HG has endpoints at point $H(7,5)$ and at point $G(11, y)$, with a length of $4\sqrt{5}$ units. What is the value of y ?
3. Line segment HK has endpoints at point $H(9,3)$ and point $K(12,-6)$. What is the length of $\overline{HK}$?